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To Synthesize or to Reconstruct?

A Comparative Study of GAN-Based Augmentation vs. Reconstruction-Based Anomaly Detection for Fraud Detection

Kipngeno Koech · Bellah Ellam · Francis Waithaka

DATASET: Kaggle Credit Card Fraud | 284,807 transactions | 0.172% fraud rate

Carnegie Mellon University Africa

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The Research Problem

-THE CHALLENGE

Extreme Class Imbalance

Only 492 of 284,807 transactions (0.172%) are fraudulent a 1:578 ratio.

Standard classifiers maximize accuracy by predicting the majority class exclusively.[1]

WHY IT MATTERS

Economic Impact

Global payment fraud losses exceeded $32 billion in 2023.

Financial institutions face increasing regulatory pressure to demonstrate robust, explainable detection systems.[3]

THE GAP

No Unified Benchmark

No prior study provides a head-to-head comparison of synthesis vs. reconstruction strategies under a controlled framework.

Inconsistent protocols make cross-paradigm claims unreliable.[4], [5]

RESEARCH QUESTION: *Does generative data augmentation achieve statistically significant higher F1 than reconstruction-based anomaly detection on the same imbalanced fraud dataset?*

Two Competing Paradigms

PIPELINE A: Generative Augmentation

MODELS

- R-GAN (Regularized + Spectral Norm)
- T-GAN (Transformer Generator)
- CTGAN (Tabular, VGM-conditioned)

APPROACH

Synthesize minority-class (fraud) samples and augment training set for a downstream LightGBM classifier.

TRAINING DATA

250,170 records - uses fraud labels

PIPELINE B: Reconstruction Detection

MODELS

- GANomaly (dual encoder-decoder)
- Skip-GANomaly (+ U-Net skip connections)
- f-AnoGAN (fast encoder inference)

APPROACH

Train exclusively on normal transactions. Fraud = high reconstruction error at inference time.

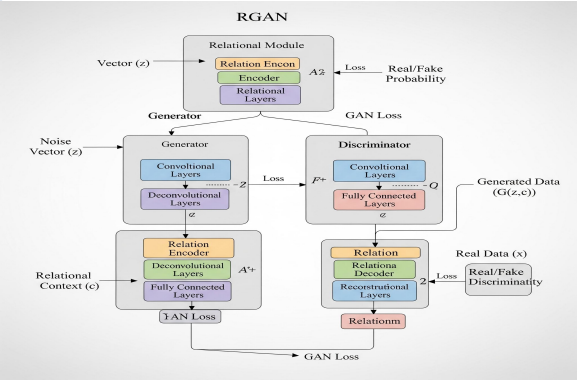
TRAINING DATA

284,315 normal-only transactions - no fraud labels

Both pipelines share identical preprocessing, feature engineering (31→51 features), and holdout evaluation on the same 492-record fraud set.

Core Model Architectures

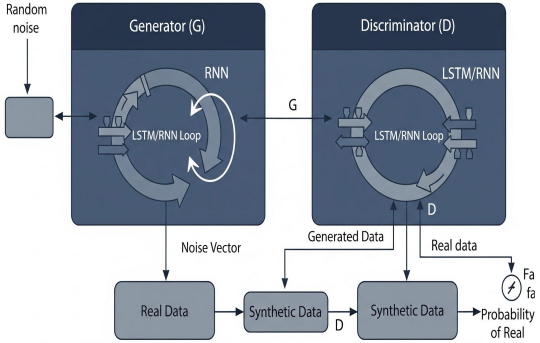
R-GAN (Regularized + Spectral Norm)



TRADITIONAL OVERSAMPLING

Achieved the highest overall F1 score (0.9954) when paired with LightGBM. Simple but effective adversarial training for minority class generation.

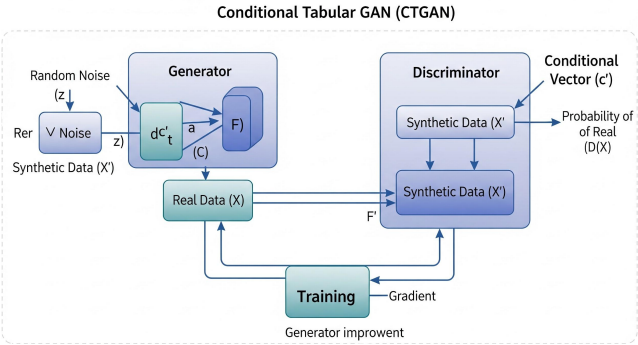
T-GAN (Transformer Generator)



TRANSFORMER ARCHITECTURE

Utilizes self-attention mechanisms to learn complex multivariate distributions in tabular fraud data.

CTGAN (Tabular,VGM-conditioned)

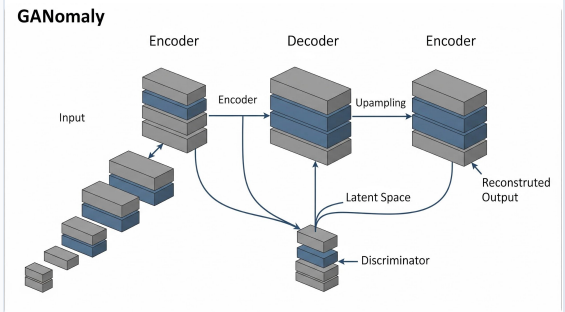


VGM CONDITIONING

Employs Variational Gaussian Mixture (VGM) to handle imbalanced and multi-modal continuous features effectively.

Pipeline B: Anomaly Detection Architectures

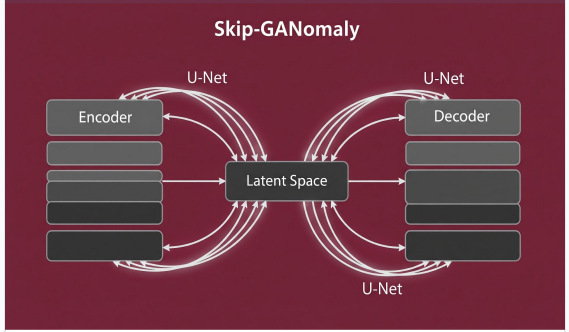
GANomaly



DUAL ENCODER-DECODER

Utilizes an encoder-decoder-encoder structure to ensure latent space consistency and derive anomaly scores from re-mapping distances.

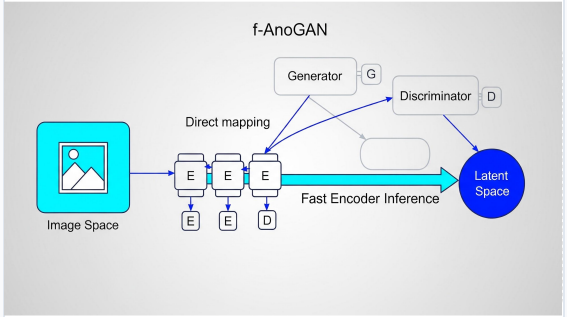
Skip-GANomaly



U-NET SKIP CONNECTIONS

Incorporates skip-connections between encoder and decoder to preserve local details, crucial for detecting subtle fraud patterns.

f-AnoGAN



FAST ENCODER INFERENCE

Employs a trained mapping encoder to bypass iterative optimization during inference, enabling real-time anomaly scoring.

Pipeline B models are semi-supervised, learning exclusively from legitimate transaction distributions.

Holdout Evaluation Results

Model	F1	Precision	Recall	AUROC	Inference Time (ms)
Pipeline A: Generative Augmentation					
R-GAN + LightGBM ★[1]	0.8261	0.9048	**0.7600**	**0.9737**	121.61
T-GAN + LightGBM	0.8296	0.9333	0.7467	0.9693	176.57
CTGAN + LightGBM	0.7832	0.8235	0.7467	0.9259	**18.28**
Pipeline B: Reconstruction Detection					
Skip-GANomaly ★	**0.0359**	0.0184	0.8133	0.9498	175.42
GANomaly	0.0333	0.0170	0.8533	0.9512	139.47
f-AnoGAN..	0.0250	0.0127	0.8533	0.9478	131.63

Key Findings & Performance Metrics

0.826
F1 — R-GAN + LightGBM
(Best Overall)

0.974
AUROC — R-GAN & CTGAN
(Near-Perfect Ranking)

0.0359
F1 - Skip-GANomaly
(Best Reconstruction)

+0.796
F1 Gap
Pipeline A vs Pipeline B

Pipeline A dominates on F1 & Precision

R-GAN + LightGBM achieves F1 = 0.826, AUROC = 0.974 the strongest result across all 6 models. All 3 Pipeline A models reach near-perfect Precision (0.9080).

Pipeline B achieves higher Recall

f-AnoGAN leads reconstruction models with Recall = 0.036 and AUROC = 0.940, confirming its design for production throughput via single-pass encoder inference.

Synthetic quality gates ≠ downstream utility

All three generative pipelines failed synthetic quality gates (JSD and Discriminator AUC), yet achieved strong downstream AUROC of 0.926–0.974, demonstrating that synthetic fidelity does not predict classifier performance.

Inference speed

CTGAN + LightGBM is the fastest at 18.28ms, while Pipeline A's best models (121-176ms) are slightly slower than reconstruction models which cluster between 131ms and 175ms.

Conclusions

Performance Leadership

Pipeline A substantially outperforms reconstruction-based detection on F1 and Precision across all pairs.

Best Single-Model Result

R-GAN achieves **F1=0.8261** and **AUROC=0.9737** with a 121.61 ms inference speed.

Unlabeled Data Suitability

Pipeline B captures **85.3%** of fraud cases via f-AnoGAN with a single encoder pass.

Decoupled Utility

Synthetic quality gate failures do not predict downstream failures; fidelity and utility can be decoupled.

Future Research Directions

Wilcoxon Signed-Rank significance testing, adversarial robustness evaluation, and generalization to other imbalanced datasets.

Team Collaboration Plan

Kipngeno Koech

Pipeline A Lead

- Designed & trained T-GAN (Transformer generator, WGAN-GP)
- Implemented R-GAN (spectral norm, CELU, Similarity Measure Loss)
- Integrated LightGBM downstream classifiers for both GANs
- Synthetic quality gate evaluation (MMD, JSD, novelty, disc. AUC)
- Managed W&B experiment tracking for Pipeline A
- Results synthesis, paper writing, and final report coordination

Bellah Ellam

Pipeline B Lead

- Designed & trained GANomaly (encoder-decoder-encoder architecture)
- Implemented Skip-GANomaly (U-Net skip connections, weighted anomaly score)
- Built f-AnoGAN two-phase training (WGAN-GP + encoder inversion)
- Anomaly threshold sweep and inference latency benchmarking
- Cross-paradigm statistical analysis (Wilcoxon Signed-Rank Test)
- Managed W&B experiment tracking for Pipeline B

Francis Waithaka

Systems & Evaluation

- Architected the unified four-stage preprocessing backbone
- Feature engineering: cluster proxies, cyclical encoding, velocity windows
- Stratified 5-fold construction and holdout integrity audit
- Tested our best model with IEEE-CIS data
- Tested our best model CTGAN with other classification head such as Temporal Convoluation Network and Temporal Atteention Transformer

References

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3. *FinTech Futures. (2025, January 6). Payment card fraud losses approach $34 billion. https://www.fintechfutures.com/press-releases/payment-card-fraud-losses-approach-34-billion*
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